4-3.

$n_0 \sin\theta_0 = n_1 \sin\theta_1 = n_2 \sin\theta_2 = C$.

$n_2 \sin\theta_2 = C = n_0 \sin\theta_0$

弧长 $(ds)^2 = (dy)^2 + (dz)^2$

$\frac{dy}{ds} = \sin\theta$.

$\therefore \left(\frac{dz}{dy}\right)^2 = \left(\frac{ds}{dy}\right)^2 - 1 = \frac{1}{\sin^2\theta} - 1 = \frac{n_2^2}{n_0^2 \sin^2\theta_0} - 1 = \frac{n_0^2 e^{2az}}{n_0^2 \sin^2\theta_0} - 1$

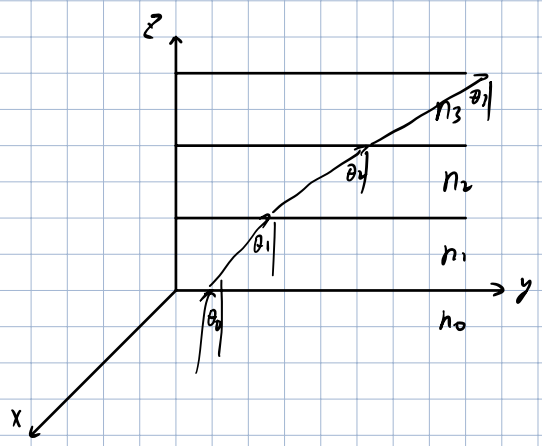
$= \frac{e^{2az}}{\sin^2\theta_0} - 1$

傍轴近似 $\theta_0 \approx 90^\circ$ $\sin\theta_0 \approx 1$. ...

$\because 0 < a << 1$ $e^{2az} \approx 1 + 2az$ $\frac{dz}{dy} = \sqrt{2az}$ $\frac{dy}{dz} = \frac{1}{\sqrt{2az}}$

$y(z) = \int \frac{1}{\sqrt{2az}} dz = \sqrt{\frac{2z}{a}} + C. = \sqrt{\frac{2z}{a}} + y_0$

同理, $x(z) = \sqrt{\frac{2z}{a}} + x_0$



4-4.